

iFOOD CRM Data Analys

Campaign Profit Optimization Model

From mass targeting to data-driven precision marketing

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Data Science | Marketing Analytics

-\$3,046

Previous Campaign Profit

Why Did the Last Campaign Fail?

Low Response Rate

Only ~15% of customers responded to the previous campaign, meaning 85% of campaign spend was wasted on non-responders.

Negative Profit

Total campaign cost: \$6,720
Total revenue: \$3,674
Resulting in a loss of \$3,046 — a negative ROI campaign.

Mass Targeting

All 2,240 customers were contacted indiscriminately, with no data-driven selection criteria to identify likely responders.



Goal: Use customer data, segmentation, and predictive modeling to improve targeting precision and maximize campaign profitability.

Understanding the Data

2,240

Total Customers

Demographics

Age & Education
Income
Marital Status
Children at Home

Spending

Wines, Fruits, Meat
Fish, Sweets, Gold
Total & Avg Spend

Channels & Engagement

Web, Catalog, Store
Web Visits/Month
Deal Purchases



Target Variable

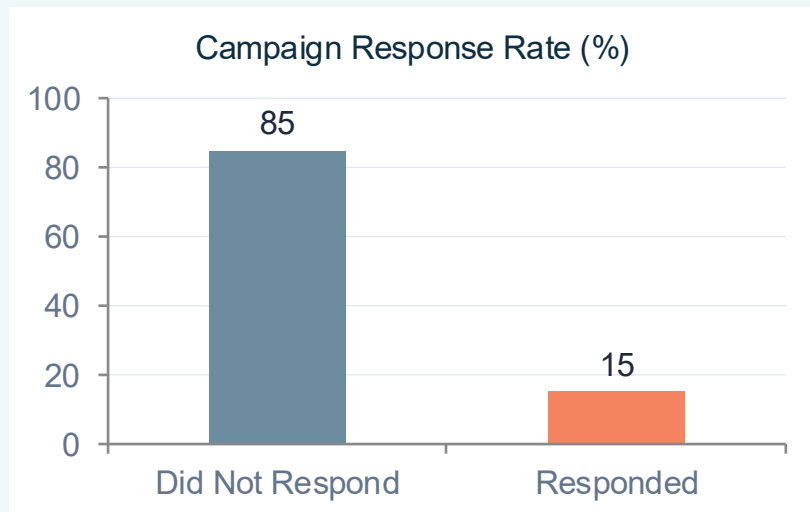
Response (0/1) — Did the customer accept the campaign offer?



Campaign History

5 previous campaigns (Cmp1–5) + Recency of last purchase (days)

What the Data Reveals



Response Rate

~15%

Only 1 in 7 customers responded — mass targeting is wasteful



High Spenders Respond More

2×

Responders have ~2× higher average total spending vs non-responders



Recency Matters

Lower = Better

Responders purchased more recently, signaling active engagement



Channel Preference

Catalog & Web

Responders use web & catalog channels more frequently

4 Distinct Customer Segments (K-Means)



Segment 0: High-Value Loyalists

Traits: High income, high spending, low recency, catalog/web buyers

Strategy: Premium offers, early access, loyalty rewards



Segment 1: Recent Active Buyers

Traits: Moderate income, recent purchases, active web engagement

Strategy: Personalized recommendations, cross-sell campaigns



Segment 2: Price-Sensitive Shoppers

Traits: Lower income, deal-driven, higher kids at home

Strategy: Discount-led promotions, value bundle campaigns



Segment 3: Dormant / Low-Engagement

Traits: High recency (long absent), low spending, low purchase frequency

Strategy: Light re-engagement touch or exclude to save cost

Predicting Who Will Respond

Model Performance Comparison

Metric	Logistic Regression	Random Forest ✓
Accuracy	~80%	~88%
Precision	~54%	~72%
Recall	~66%	~61%
F1 Score	~59%	~66%
ROC AUC	~0.83	~0.91

⚠ Class imbalance handled via stratified split + `class_weight='balanced'`. Accuracy alone is misleading — ROC AUC and F1 are the true measures of success.



Winner

Random Forest

ROC AUC: 0.91

Better precision on imbalanced class

Used for profit threshold optimization

Threshold-Based Targeting Strategy

Estimated Profit by Probability Threshold



Before: 2,240 contacted | Response: 15% | Profit: -\$3,046

After: ~400–600 contacted | Response: ~35–45% | Profit: +\$700–900

1

Score Customers

Random Forest assigns each customer a response probability (0–1)

2

Set Threshold

Only contact customers above the chosen probability threshold

3

Optimize Profit

Select threshold that maximizes: Revenue – Campaign Cost

Action Plan for iFood



Target High-Probability Customers

Use Random Forest scores to contact only customers above the optimal threshold. Reduces cost and focuses spend where it converts.



Prioritize High-Value Segments

Segment 0 (High-Value Loyalists) and Segment 1 (Recent Active Buyers) yield the best ROI. Allocate budget here first.



Personalize by Segment

Send premium offers to loyalists, product recommendations to active buyers, and discount bundles to price-sensitive shoppers.



Reduce Spending on Low-Response Segments

Dormant customers (Segment 3) drain budget with minimal return. Use low-cost re-engagement emails or exclude entirely.

Thank You

iFood Campaign Profit Optimization | Nicholas Hartowidjojo

2,240

Customers Analyzed

4

Segments Identified

91%

Model ROC AUC

+Profit

vs -\$3,046 Before